How long must an antibody trial run?

Methods and results from the simulation dry run

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Bottom line. Under a correctly specified biphasic model, six months of monthly follow-up with $N = 100$ was the first adequate dry-run design, even though the true threshold crossing occurred at 396 days. Longer follow-up was more valuable than a larger cohort for learning the tail. The main threat was model form: plausible misspecified models produced narrow, badly biased intervals or a high probability of predicting that the threshold was never crossed.

Executive summary

The study asks when a decaying vaccine-induced antibody marker falls below a protection-relevant threshold. The duration estimand is

$$T^* = \inf\{t \geq 0 : c(t) \leq c_{\text{thr}}\}.$$

The dry run simulated complete trajectories, hid observations after each candidate review date, fit alternative decay families, extrapolated to T^*, and compared the inferred duration with the known truth.

The well-specified biphasic fit was nearly unbiased throughout. At three months its interval was honestly enormous; at six months, after the fast-to-slow transition became visible, relative interval width fell below the 0.50 adequacy target. More participants reduced noise, but extra follow-up revealed the slow tail more efficiently. Misspecification was much more serious: exponential fits predicted loss of protection too early, power-law fits often too late, and plateau fits assigned substantial mass to $T^* = \infty$.

Full NUTS was important. A previous curvature-based Laplace approximation understated uncertainty and altered conclusions about both adequacy and model selection. The hierarchical extension also suggested a substantive distinction: increasing N improved the population-duration estimate, whereas individual durations remained governed mainly by within-person follow-up. That extension is provisional because some fits had severe convergence problems.

1 Methods

1.1 Scientific setup

The primary truth was a two-component exponential decline,

$$c(t) = (1 - f)c_0 e^{-k_1 t} + f c_0 e^{-k_2 t}, \qquad k_1 > k_2 > 0,$$

with multiplicative log-normal measurement noise. A richer mechanistic ODE truth provided a robustness check. The test-run calibration used $c_0 = 5000$, $f = 0.30$, fast half-life 14 days, slow half-life 250 days, $\sigma_{\log} = 0.20$, and $c_{\text{thr}} = 500$. This gave a true T^* of 396 days (about 13 months).

The fixed threshold represented approximately 50% protection from symptomatic infection. The method also allowed a logistic marker-to-protection map, but the reported dry-run results focused on the threshold-crossing estimand.

Component	Dry-run implementation	Purpose
Truth	Biphasic exponential; mechanistic ODE robustness truth	Known long-term trajectory
Observations	Log-normal noise; monthly visits in the NUTS re-run	Mimic repeated antibody measurements
Fitted families	Exponential, biphasic, power-law and plateau	Measure model-form sensitivity
Inference	Full NUTS; 1000 warm-up + 1000 draws, two chains	Propagate parameter uncertainty into T^*
Follow-up	3, 6, 9, 12 and 18 months	Reveal how much tail observation is needed
Cohort size	$N = 50, 100, 200$ in the homogeneous comparison	Contrast participants with follow-up
Primary fidelity	60 Monte Carlo replicates on the main grid	Estimate bias, coverage and precision

1.2 Inference and evaluation

Each posterior parameter draw was transformed into a threshold-crossing time. A trajectory that did not cross the threshold retained $T^* = \infty$ rather than being truncated to an arbitrary finite value. The main summaries were bias, empirical interval coverage, relative interval width, and the fraction of non-crossing draws. Model-selection frequency and AIC-weighted model averages were also examined.

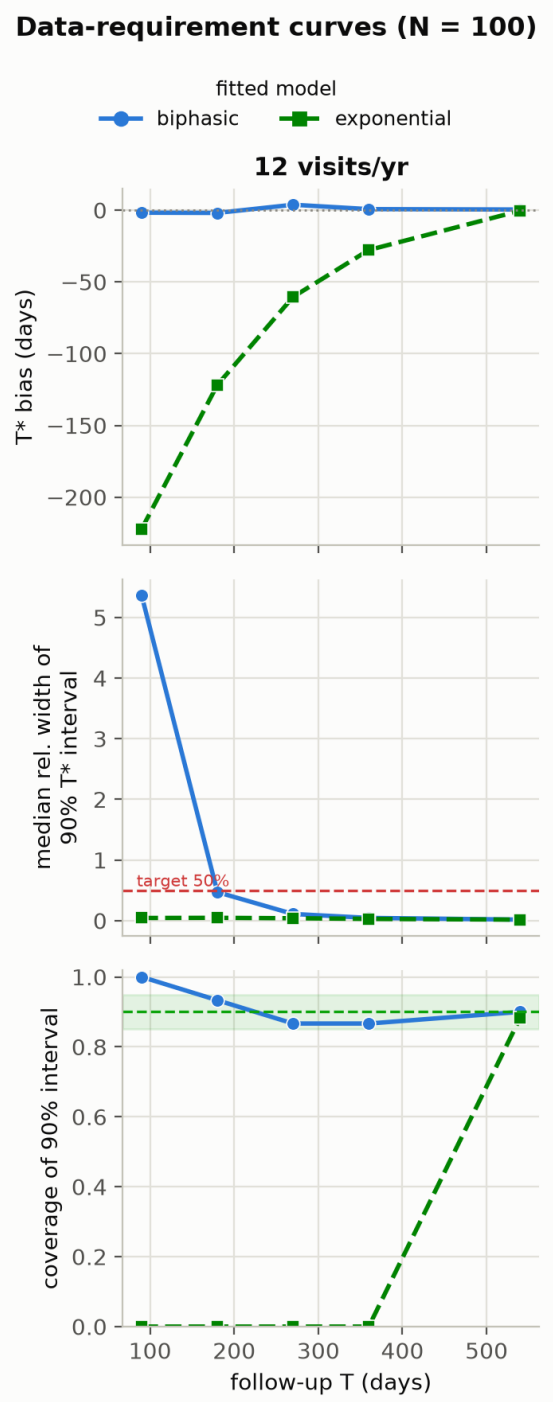
A design was treated as adequate when its interval coverage was within the allowed tolerance, its median relative width was at most 0.50, and non-crossing predictions did not dominate. This joint rule prevents a very wide interval from passing on coverage alone and prevents a narrow but biased interval from appearing successful.

2 Results

2.1 Correct model: precision, not bias, was the constraint

For the correct biphasic fit at $N = 100$ with monthly visits, median bias stayed between -2 and $+4$ days. The major change was precision: relative width fell from 5.36 at three months to 0.47 at six months and 0.11 at nine months. The six-month cell was therefore the first adequate design in the dry run.

Follow-up	3 mo	6 mo	9 mo	12 mo	18 mo
Coverage	1.00	0.93	0.87	0.87	0.90
Median relative width	5.36	0.47	0.11	0.045	0.017
Median bias (days)	-2	-2	$+4$	$+1$	0
Adequate	no	yes	yes	yes	yes

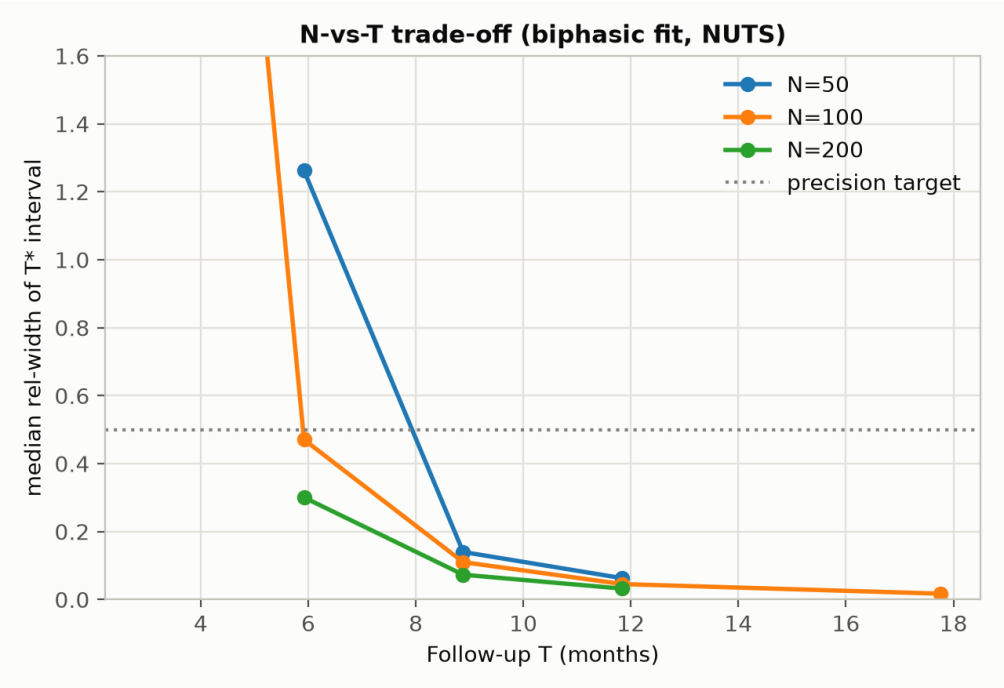


Bias, relative interval width and coverage versus follow-up at $N = 100$ with monthly visits.

The sharp gain between three and six months coincided with observation beyond the fast/slow transition. Coverage of 1.00 at three months was not evidence of usefulness: it arose from an interval more than five times the true duration.

2.2 Follow-up dominated cohort size in the homogeneous analysis

At six months, median relative width was 1.26, 0.47 and 0.30 for $N = 50, 100, 200$. At nine months it was 0.14, 0.11 and 0.07. Thus $N = 50$ at nine months was about twice as precise as $N = 200$ at six months. More participants tightened inference around the observed trajectory, but could not replace seeing the slow tail.

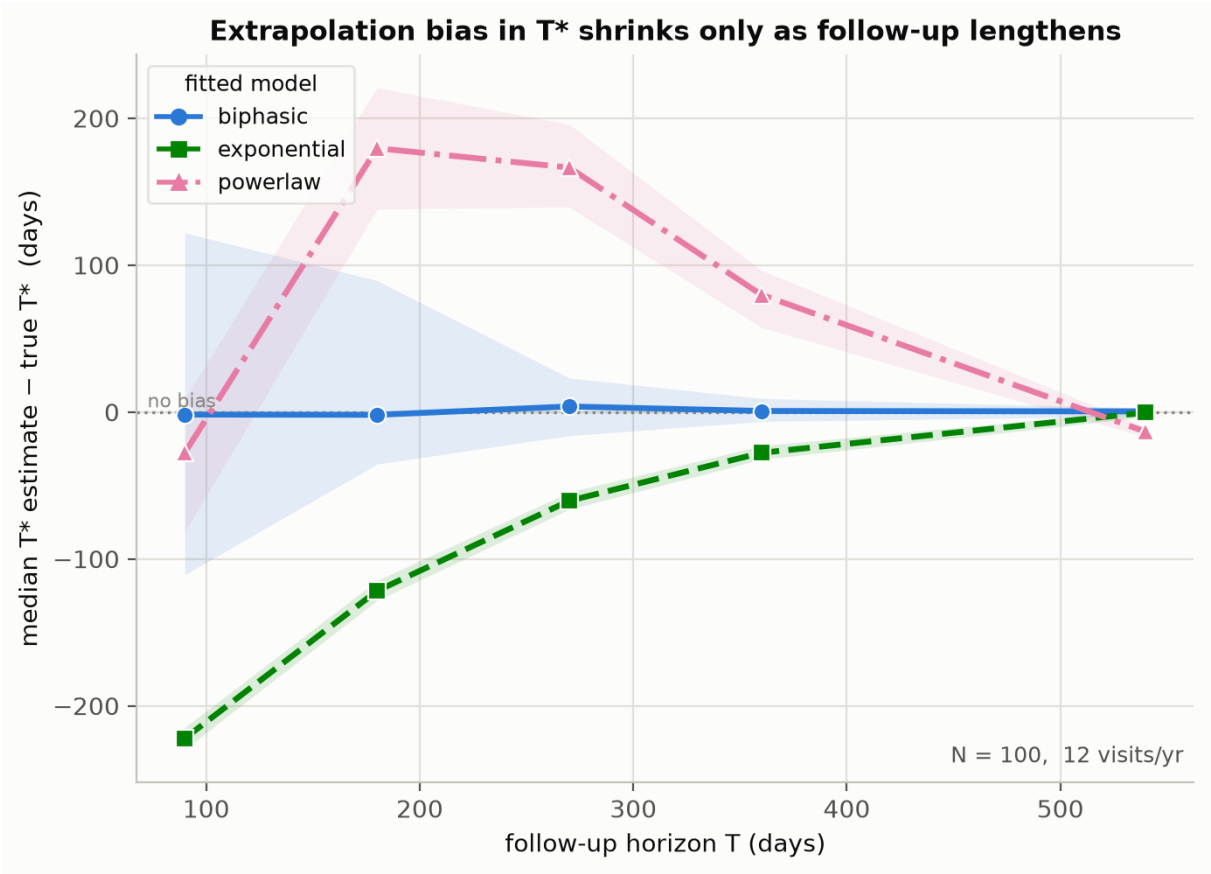


Participant number helped, but three additional months of follow-up beat a four-fold cohort increase in the highlighted comparison.

2.3 Model misspecification was the headline risk

The direction of error depended on the fitted family. Exponential fits extrapolated the early decline and predicted threshold crossing too soon; power-law fits often predicted it too late. Plateau fits frequently interpreted the slow tail as a floor above the threshold.

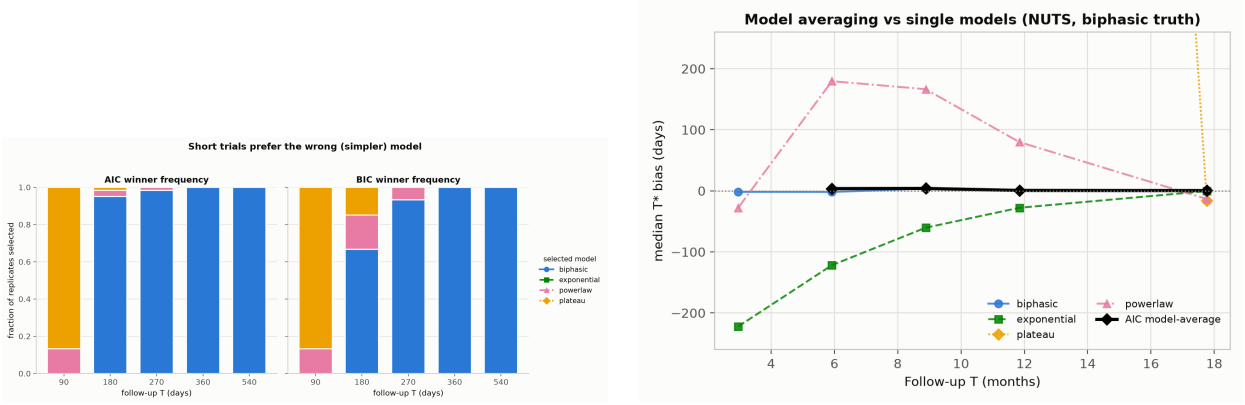
Median bias (days)	3 mo	6 mo	9 mo	12 mo	18 mo
Biphasic	−2	−2	+4	+1	0
Exponential	−222	−122	−60	−28	0
Power-law	−28	+180	+166	+80	−13
Plateau	predominantly never crossing				−16



Median duration bias by fitted family. The correct biphasic fit remained near zero; misspecified models failed in different directions.

The exponential interval was especially misleading: its relative width was about 0.04, yet it had zero coverage until 18 months. The power-law fit also had zero coverage from six months onward. For the plateau family, 97–100% of posterior draws failed to cross the threshold through 12 months.

Model selection helped only after sufficient follow-up. At three months AIC selected the plateau model 87% of the time. At six months it selected the correct biphasic family 95% of the time (BIC: 67%), and by 12 months selection was effectively certain. AIC-weighted averaging inherited the plateau failure at three months and produced an infinite median duration; from six months onwards its median bias was only about 0.4–3.9 days.



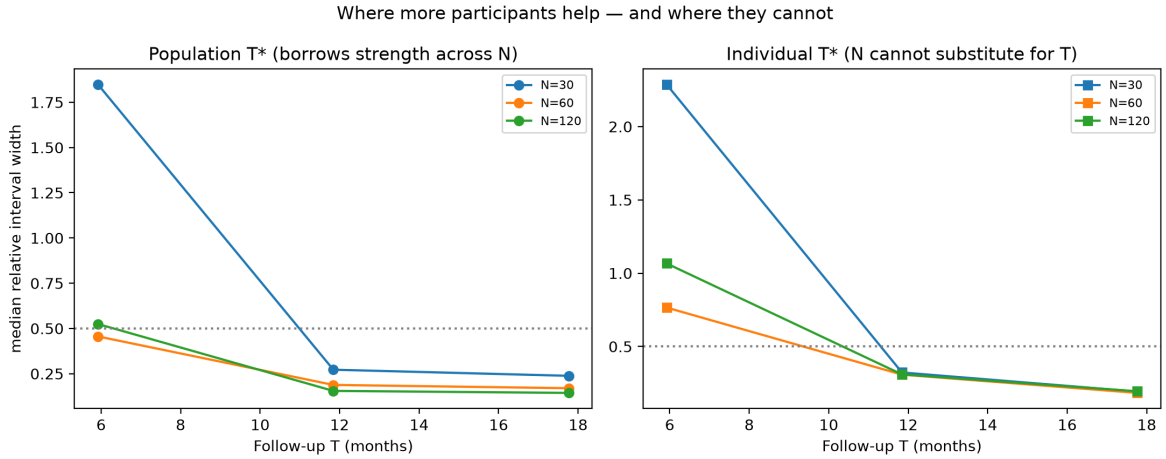
Model selection recovered after the transition was observed; model averaging did not rescue the shortest follow-up.

2.4 Robustness to a richer truth

When the biphasic model was fit to data generated by the three-timescale mechanistic ODE, median bias fell from +173 days at three months to +15, +3, +1 and −1 days at 6, 9, 12 and 18 months. Coverage was 1.00, 0.83, 0.83, 0.87 and 0.93. The main conclusion survived, although the modest under-coverage at intermediate follow-up showed that even a close mechanistic mismatch mattered.

2.5 Heterogeneity separated population and individual questions

The hierarchical extension used log-normal random effects on peak and slow rate with correlation -0.3, and compared $N = 30, 60, 120$ at 6, 12 and 18 months. At 12 months, population relative width improved from 0.27 to 0.19 to 0.15 as N increased. Individual relative width remained approximately 0.31–0.32 at all three cohort sizes. Larger cohorts therefore bought information about the population distribution, but not equivalent information about each person's unobserved tail.

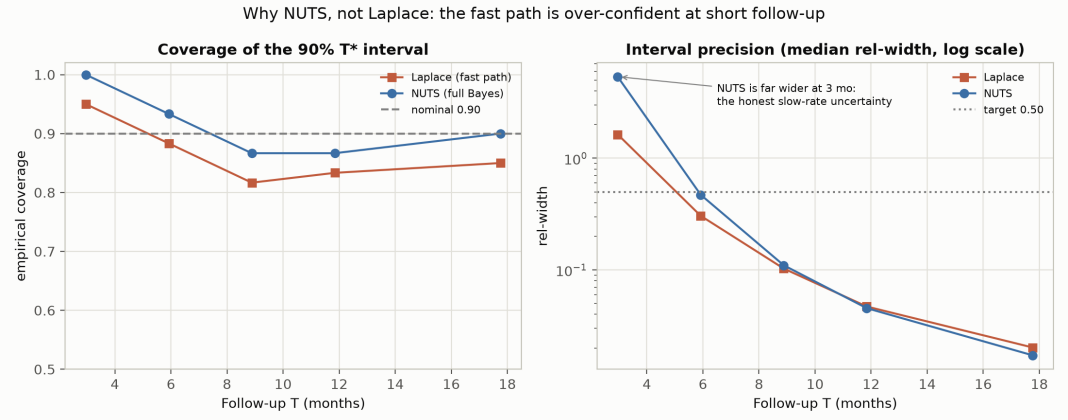


Population and individual durations responded differently to increasing participant number.

This finding is provisional. Several hierarchical replicates had maximum $\widehat{R}$ as high as about 1.85 despite 1200 tuning steps and a target acceptance probability of 0.95. Population coverage was about 0.83 at six months and conservative at later reviews; individual coverage was approximately 0.83–0.91. The qualitative N-versus-follow-up pattern is useful, but individual cells should not be treated as production estimates.

2.6 Why the full posterior calculation mattered

The earlier MAP-plus-Laplace path under-covered at every follow-up (about 0.82–0.85 versus 0.87–1.00 under NUTS). At three months it reported relative width 1.63 where NUTS reported 5.36. It also made adequacy non-monotonic and delayed apparent recovery of the correct model family to about 12 months because the optimiser under-fit the biphasic likelihood. Full NUTS changed both the uncertainty and model-selection conclusions.



Laplace approximations understated uncertainty and distorted the apparent data requirement.

3 Interpretation and limitations

The dry run supports three claims. First, observation beyond the kinetic transition can be more valuable than a large increase in cohort size. Second, apparently precise extrapolation is not trustworthy unless model form is tested. Third, population and individual duration are distinct estimands with different data requirements.

The numerical thresholds remain preliminary. Sixty replicates give a coverage standard error of roughly 0.04 near the target, so boundary decisions are noisy. The NUTS re-run used monthly visits only. Threshold, censoring and noise sensitivities were not swept; the severe-disease duration lay beyond all observed design cells. The short-tail test calibration also made extrapolation easier than the long-tail baseline now proposed. Finally, AIC and BIC were reconstructed from a profile likelihood; LOO or stacking would be preferable for a fully Bayesian comparison.

4 Differences from the current master plan

The dry run was a feasibility exercise, not an execution of the current workshop plan. The most important differences are summarised below.

Area	Dry run reported here	Current master plan
Calibration	Fast/slow half-lives 14/250 days and long-lived fraction 0.30; true $T^* = 396$ days	Starts from the handout's tentative 35/581-day half-lives, 75-day switch and about 80:20 weighting; requires explicit slow-tail, fraction and threshold sensitivities
Baseline grid	3–18 months in the reported NUTS grid; monthly visits; $N = 50, 100, 200$ comparisons	Includes 24 months and both monthly and quarterly visits; baseline $N = 100$, followed by $N = 30, 100, 300$ sensitivity cells
Sampling frequency	The full-NUTS results fixed frequency at monthly; quarterly conclusions came from the earlier approximation	Frequency is a planned baseline design axis, with monthly and quarterly schedules scored using NUTS
Sampler fidelity	Main grid used two chains with 1000 warm-up and 1000 retained draws	Final results should use four chains with at least 1000 warm-up and 1000 retained draws; two chains are development fidelity only
Hierarchical fits	Two chains, 1200 tuning and 800 retained draws; some $\widehat{R}$ values reached about 1.85	At least 1500 warm-up iterations, target acceptance near 0.95, and a pre-agreed convergence filter before aggregation
Adequacy reporting	Used the 0.50 relative-width target and a non-crossing guard; did not report a systematic stricter-width analysis	Requires coverage ≥ 0.85, median relative width ≤ 0.50, non-crossing probability ≤ 0.10, plus repetition at a 0.30 width target
Sensitivities	Threshold, censoring, noise and long-tail calibration were not swept	Treats threshold and assay limits as first-class configuration fields and retains explicit calibration sensitivities before headline claims
Production status	Sixty main-grid replicates; hierarchical cells used six replicates and include unconverged fits	About 50–60 replicates are workshop-development fidelity; important cells should later rise to about 200 with Monte Carlo uncertainty and diagnostic filtering
Model uncertainty	Added AIC-weighted averaging; at three months its median was infinite	The protected Phase 2 deliverable is a diagnosed family comparison and model-selection result; model averaging is not required for the workshop core

Practical implication. The dry run justifies the master plan's safeguards—full NUTS for scored cells, staged grids, explicit non-crossing summaries, schema-first integration and strict diagnostics—but its numerical minimum-viable design should not be carried forward as a final recommendation until the current calibration, frequency grid and convergence policy have been applied.

Prepared as a synthesis of the archived methods specification and archived test-run results, with a final comparison against the current coordinator plan. No model was refit and no result was recomputed.